

## 2 Amnesiac Systems in FreeBSD

In this technical note, we explore configurations of the FreeBSD operating system. With a goal of creating an amnesiac system that keeps the operating system separate from working storage drives, we experiment with various configurations. We demonstrate the planning of drive index configuration coding and the reasoning behind the choices. First, we code a system that uses the operating system on the storage drive discs. Then we evolve the plan into using a live cd for operating system use with drives that are connected to the immutable system in Random Access Memory (RAM) via configuration code. Using an amensiac system like this, we would be able to reset the operating system to a known immutable state with every startup. The drives could be disconnected and reattached to other systems for various purposes. Throughout, we cover specific commands for drive and operating system management and operation.

### 2.1 Introduction

#### 2.1.1 Background and Content

In order to maintain persistence, download files, or interact with a file system, attackers need to be able to write to a system. Many active attack techniques require some kind of writing action as a precedent for execution. Based on our limited penetration testing and attack experience, we identified 60 Enterprise Technique groups in MITRE ATT&CK [1] that could be thwarted by removing an attacker's ability to write. This listing is not comprehensive or absolute; we reviewed these technique categories and recognized that at least a naive attempt to carry out those techniques might involve writing to a file or process. We ignored more passive reconnaissance techniques and those that would be executed outside of the target system itself. Still, we were able to generate a formidable list of attack techniques that would at least be hampered by an operating system that was designed to be immutable.

We cover our listing of these MITRE ATT&CK techniques in Tables 2.1, 2.2, 2.3, 2.4, 2.5, and 2.6. As we review the list, we can see how some might fall into the trap of thinking such a system was invincible or "hack proof". Our past experience with attack and defense warns us to accept these extensive defenses as tentative. An ingenious attacker might find a way to carry out these techniques without meeting our preconceived ideas that they require writing or that our chosen forms of immutability would protect against that writing. Nonetheless, the lengthy list of major groups of techniques and their longer list of subtechniques implies that immutability in the right ways can serve as a provocative and solid form of defense.

#### 2.1.2 Problem Statement

Our goal is to keep the core services and operating system stable and immutable through physical security controls. This will allow the system to resist common

Table 2.1: MITRE ATT&CK Techniques Require Writing

| Technique Group                 | Concept                                                                                                                                                                  |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Create Account                  | "Adversaries may create an account to maintain access to victim systems."[2]                                                                                             |
| Create or Modify System Process | "Adversaries may create or modify system-level processes to repeatedly execute malicious payloads as part of persistence."[3]                                            |
| Data Destruction                | "Adversaries may destroy data and files on specific systems or in large numbers on a network to interrupt availability to systems, services, and network resources. "[4] |
| Data Encoding                   | "Adversaries may encode data to make the content of command and control traffic more difficult to detect."[5]                                                            |
| Data Manipulation               | "Adversaries may insert, delete, or manipulate data in order to influence external outcomes or hide activity, thus threatening the integrity of the data."[6]            |
| Data Obfuscation                | "Adversaries may obfuscate command and control traffic to make it more difficult to detect."[7]                                                                          |
| Data Staged                     | "Adversaries may stage collected data in a central location or directory prior to Exfiltration."[8]                                                                      |
| Defacement                      | "Adversaries may modify visual content available internally or externally to an enterprise network, thus affecting the integrity of the original content."[9]            |
| Deploy Container                | "Adversaries may deploy a container into an environment to facilitate execution or evade defenses."[10]                                                                  |
| Develop Capabilities            | "Adversaries may develop capabilities that can be used during targeting."[11]                                                                                            |

Table 2.2: MITRE ATT&amp;CK Techniques Require Writing

| Technique Group                             | Concept                                                                                                                                                                                                                                                                           |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Disable or Modify System Firewall           | "Adversaries may disable or modify host-based or network firewalls to impair defensive mechanisms and enable further action." [12]                                                                                                                                                |
| Disable or Modify Tools                     | "Adversaries may disable, degrade, or tamper with security tools or applications (e.g., endpoint detection and response (EDR) tools, intrusion detection systems (IDS), antivirus, logging agents, sensors, etc.) to impair or reduce visibility of defensive capabilities." [13] |
| Disk Wipe                                   | "Adversaries may impair the availability of systems and data by destroying, corrupting, or overwriting data." [14]                                                                                                                                                                |
| Domain or Tenant Policy Modification        | "Adversaries may modify the configuration settings of a domain or identity tenant to evade defenses and/or escalate privileges." [15]                                                                                                                                             |
| Escape to Host                              | "Adversaries may break out of a container to gain access to the underlying host." [16]                                                                                                                                                                                            |
| Event Triggered Execution                   | "Adversaries may establish persistence and/or elevate privileges using system mechanisms that trigger execution based on specific events." [17]                                                                                                                                   |
| Exclusive Control                           | "Adversaries who successfully compromise a system may attempt to maintain persistence by "closing the door" behind them – in other words, by preventing other threat actors from initially accessing or maintaining a foothold on the same system." [18]                          |
| Execution Guardrails                        | "Adversaries may use execution guardrails to constrain execution to specific environments." [19]                                                                                                                                                                                  |
| Exfiltration Over Physical Medium           | "Adversaries may exfiltrate data using removable media." [20]                                                                                                                                                                                                                     |
| File and Directory Permissions Modification | "Adversaries may modify file and directory permissions or attributes to evade access control restrictions." [21]                                                                                                                                                                  |

Table 2.3: MITRE ATT&CK Techniques Require Writing

| Technique Group               | Concept                                                                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Firmware Corruption           | "Adversaries may use malicious code embedded in firmware to maintain persistence or disrupt systems."[22]                                        |
| Hide Artifacts                | "Adversaries may abuse functionality to hide artifacts from the user or security tools."[23]                                                     |
| Hijack Execution Flow         | "Adversaries may execute their own malicious payloads by hijacking the way operating systems load programs."[24]                                 |
| Implant Internal Image        | "Adversaries may implant malicious software into software images to establish persistence."[25]                                                  |
| Indicator Removal             | "Adversaries may delete or modify artifacts generated on a system to remove evidence of their presence."[26]                                     |
| Ingress Tool Transfer         | "Adversaries may transfer tools or other files from an external system into a compromised environment."[27]                                      |
| Inhibit System Recovery       | "Adversaries may interrupt availability of system and network resources by inhibiting system recovery."[28]                                      |
| Lateral Tool Transfer         | "Adversaries may transfer tools or other files between systems in a compromised environment."[29]                                                |
| Masquerading                  | "Adversaries may attempt to manipulate features of their artifacts to make them appear legitimate or benign to users and/or security tools."[30] |
| Modify Authentication Process | "Adversaries may modify authentication processes and mechanisms to bypass or weaken authentication requirements."[31]                            |

Table 2.4: MITRE ATT&CK Techniques Require Writing

| Technique Group                     | Concept                                                                                                                                                            |
|-------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modify Cloud Compute Infrastructure | "Adversaries may create, modify, or delete cloud resources to achieve their objectives."[32]                                                                       |
| Modify Cloud Resource Hierarchy     | "Adversaries may attempt to modify hierarchical structures in infrastructure-as-a-service (IaaS) environments in order to evade defenses."[33]                     |
| Modify Registry                     | "Adversaries may interact with the Windows Registry to hide configuration information, modify system settings, or remove information as part of their attack."[34] |
| Modify System Image                 | "Adversaries may modify the configuration of network devices to maintain access, modify traffic flows, or disrupt services."[35]                                   |
| Multi-Stage Channels                | "Adversaries may use multiple command and control channels to communicate with compromised systems."[36]                                                           |
| Obfuscated Files or Information     | "Adversaries may obfuscate their malicious code, configuration files, or other artifacts to make detection and analysis more difficult."[37]                       |
| OS Credential Dumping               | "Adversaries may attempt to dump credentials to obtain account information that can be used to access systems and services."[38]                                   |
| Plist File Modification             | "Adversaries may modify browser extensions to execute malicious code or collect information from a victim's browser."[39]                                          |
| Poisoned Pipeline Execution         | "Adversaries may manipulate continuous integration / continuous development (CI/CD) processes by injecting malicious code into the build process."[40]             |
| Power Settings                      | "Adversaries may use virtualization and containerization technologies to create or maintain access to compromised environments."[41]                               |

Table 2.5: MITRE ATT&CK Techniques Require Writing

| Technique Group                 | Concept                                                                                                                                          |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Pre-OS Boot                     | "Adversaries may abuse Pre-OS Boot mechanisms as a way to establish persistence on a system."[42]                                                |
| Prevent Command History Logging | "Adversaries may impair command history logging to hide commands they run on a compromised system"[43]                                           |
| Process Injection               | "Adversaries may inject code into processes in order to evade process-based defenses as well as possibly elevate privileges."[44]                |
| Protocol Tunneling              | "Adversaries may use protocol tunneling to hide network communications through existing protocols or services."[45]                              |
| Proxy                           | "Adversaries may use a connection proxy to direct network traffic between systems or act as an intermediary for network communications."[46]     |
| Reflective Code Loading         | "Adversaries may reflectively load code into a process in order to conceal the execution of malicious payloads."[47]                             |
| Remote Access Tools             | "An adversary may use legitimate remote access tools to establish an interactive command and control channel within a network."[48]              |
| Rogue Domain Controller         | "Adversaries may register a rogue Domain Controller to enable manipulation of Active Directory data."[49]                                        |
| Rootkit                         | "Adversaries may use rootkits to hide the presence of programs, files, network connections, services, drivers, and other system components."[50] |
| Scheduled Task/Job              | "Adversaries may abuse task scheduling functionality to facilitate initial or recurring execution of malicious code."[51]                        |

Table 2.6: MITRE ATT&amp;CK Techniques Require Writing

| Technique Group           | Concept                                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scheduled Transfer        | "Adversaries may attempt to take screen captures of the desktop to gather information over the course of an operation."[52]                                         |
| Screen Capture            | "Adversaries may abuse legitimate extensible development features of servers to establish persistent access to systems."[53]                                        |
| Server Software Component | "Adversaries may gain access to and use centralized software suites installed within an enterprise to execute commands and move laterally through the network."[54] |
| Software Deployment Tools | "Adversaries may upload, install, or otherwise set up capabilities that can be used during targeting."[55]                                                          |
| Stage Capabilities        | "Adversaries may use shared content to move malicious files or code between systems."[56]                                                                           |
| Taint Shared Content      | "Adversaries may create or modify references in user document templates to conceal malicious code or force authentication attempts."[57]                            |
| Template Injection        | "Adversaries may use traffic signaling to hide open ports or other malicious functionality used for persistence or command and control."[58]                        |
| Traffic Signaling         | "Adversaries may compromise a network device's encryption capability in order to bypass encryption that would otherwise protect data communications."[59]           |
| Weaken Encryption         | "Adversaries may bypass application control and obscure execution of code by embedding scripts inside XSL files."[60]                                               |
| XSL Script Processing     | "Adversaries may bypass application control and obscure execution of code by embedding scripts inside XSL files."[61]                                               |

attacks such as the encryption of files from ransomware and persistent account takeovers.

### 2.1.3 Project Objectives

Our system, hostname Blondie, set an example for amnesiac system implementation and design with FreeBSD. We hope that this example system will raise awareness and set patterns that will allow others to gain some of the same risk controls in their systems.

### 2.1.4 Scope of Work

Hypothesis: H1

Attackers cannot destructively interact with systems without writing to them.

Some reflection on "cannot" may lead us into conflict with this hypothesis; so, let's consider carefully what we mean. At some point in the kill chain, if an attacker wants to stop being passive and start being active and destructive, then they must somehow modify a file on the target system or write a file to the target system. That file could be in RAM; it could be in a temporary directory; it could already exist. Not all attacks require this behavior, but many do. With some imprecision we assert that attackers need to write to destroy.

ingress tool transfer MITRE ATT&CK

Hypothesis: H2

An immutable operating system will be more resistant to destructive attack.

Hypothesis: H3

An immutable operating system will promote system integrity.

With ransomware as a dominant contemporary threat, integrity of the CIA triad rises to the forefront of defensive needs.

Hypothesis: H4

An immutable operating system will promote rapid recovery.

With core components and configurations unchanged, an immutable system restarted after an attack will experience a more rapid recovery.



### 2.1.5 Variables

#### Non-Adversarial Threats

- Hardware limitations
- Coding and configuration errors
- Physical deficiencies
- Disasters

#### Adversarial Threats

- Ransomware
- Network attacks
- Physical attacks
- Non-malicious insider threats

### 2.1.6 Assumptions, Risks, and Limitations

#### Assumptions

- Sufficient time and control for power up, power down, and maintenance operations
- Direct control of server stack allows for optical disc use
- No migration of an existing commercial stack
- No commercial pressure for business performance or profit

#### Risks

- New procedures require rehearsal
- Complex research patterns may cloud known references
- Endeavor may be too complex to implement for some teams
- Unforeseen compatability or integration problems

#### Limitations

- Increased complexity requires trained personnel
- Untaught patterns will require time to learn through experimentation
- No known defense against zero day vulnerabilities in the OS
- Size constraints on optical discs may limit immutable segment of system

### 2.1.7 Significance of the Project

## 2.2 Literature Review

### 2.2.1 Overview of Existing Solutions or Approaches

#### Immutability as a Constraining Concept

We wanted to better understand immutability as an idea. We came across an article ("Immutability as an Operational Constraint Rather Than a Security Feature")[62] that pointed out that immutability can be a limiting factor. The author warned, "By just making something immutable you haven't necessarily made it safe; what you have effectively done is taken away the possibility of alteration and replaced it with a new operational reality: correction must happen through additional writes, and recovery must happen through reconciliation rather than repair"[63]

In this approach, the entire system was assumed to be an immutable system. This helped us to realize what a benefit it was to not make the entire system immutable. Our original intention was to have centralized, core, aspects of the operating system be immutable; other parts of the system could be changed within the rules and limits set by the OS.

Still, those changeable parts might have logging and monitoring activities, such as database snapshots set up in a way that allowed these cautions about immutability to contribute to our planning. Mallikarjun Vppalapati pointed out problems that could be had when records of on-system actions became immutable. Suppose a log or record was contaminated with malware or privacy information that needed to be forgotten under General Data Protection Regulation (GDPR). Administrators might need to undertake complicated edits to allow future auditors to bypass data that should not be considered. Vppalapati outlined a situation in which a simple programmatic error needed an elaborate and complicated fix because simple overwrites were not available.[64]

Vppalapati described three kinds of immutability: hard, soft, and operational. "Hard immutability describes a state of being unchangeable where changes are so heavily protected by technical mechanisms and therefore practically infeasible the alteration process is therefore practically impossible."[65] "Soft immutability...can be reverted by using privileged credentials."[66] "Operational immutability implies the lack of changes without the use of technology"[67] In our proposed system, we would use hard immutability in the optical read-only recording of the core OS. We would use operational immutability as a control because we were applying an optical disc that required physical access to the server for alterations. For example, to change the operating system burned to optical disc, an attacker or operator would need to build a new disc and physically insert it into the system, and then follow our proscribed startup sequence to get the Operating System (OS) into RAM. We would also use soft immutability, in the writing of data to files in attached storage that was not automatically loaded by the system on boot.

### Immutability Examples in Blockchains

If there's a technology that has a reputation for immutability, it is the blockchain. Simply put, the blockchain is a chain of cryptographic records; the starting point for the math used to encrypt each block is based on the previous record; the cumulative effect of the sequence of encryption is that if one record is corrupted or modified, then the break in the logic for decrypting subsequent blocks will reveal the change. So long as the records in the blockchain are accessible, users can assume that the previous entries have been unmodified. Blockchain has other characteristics; its records are usually distributed over networks and internets; but, the mathematical sequence of encryption and decryption based on preceding blocks is the source of its logical strength as an immutable record.

Blockchains, being immutable, can run into conflict with one of the more famous requirements for mutability of our time: GDPR's Right to be Forgotten (Rtbf). [68][69] One of the components of GDPR, Recital 65, says that a person who has their data recorded has the right to request a modification or deletion of that data. There are some exceptions and provisions for children who get their data recorded, but at its core Recital 65 provides every person in the EU with the legal right to have their records modified. [70] Recital 66 says that a data controller has to inform processors to erase links to or copies of personal data. This way a request to be forgotten can't just stop with one entity or be restricted to entities that the requestor discovers on their own. [71] In these laws, we don't see exceptions for immutable systems built-in. On the contrary, by being left out altogether the responsibility for dealing with these legal conflicts seems to go back on the immutable system owner. If a system were to operate that was truly immutable, then if that system contained personal data that was to be erased upon a given request: it might require the destruction of the whole system to bring the operator into legal compliance with GDPR. This might not be as catastrophic as it sounds; perhaps an operator could take a system offline, delete a record, and then completely regenerate the records with the required deletions omitted. However, multiple such mutations could lead to complexities in system maintenance with multiple required regenerations. The problem could get further complicated if the system had subsequent sequential modifications. Altogether, system operators would be cautioned not to store personal data in immutable file structures. Perhaps we might be further cautioned to keep immutable only what was operationally necessary as a safeguard against future laws or requirements. As we work through the implementation of our immutable system, we shall see that sometimes the constraints can bring us structure and sometimes the limitations are innovation accelerators.

Blockchains offer an example of Vppalapati's hard immutability and operational immutability. The European Union (EU)'s GDPR counters with an example of soft immutability requirements for certain kinds of data. We can conclude that most personal data would have to be kept off of the optical discs (and out of any network distributed blockchains). While most Personally Identifiable Information (PII) would not seem to be a part of an operating system, later we'll see that a user's full name is requested by the system as a normal part of user account setup. We could have a small conflict that leads to an involved repair if we're not careful.

We found an article which examined the types of problems with blockchains in detail. The authors classified the problems into four main categories:

Immutability Examples in Database Administration

2.2.2 Comparisons of Technologies or Frameworks

Immutability and MAC Biba

Mac biba is a mandatory access control scheme, similar to military multilevel security classifications, but with an emphasis on file integrity. `mac_biba` allows for write down and read up; this is the opposite direction from mandatory access control schemes like military security classifications which can write up and read down. [72] [73] FreeBSD offers `mac_biba` as part of its operating system; unfortunately, OpenBSD does not. [74] ... compare with diagrams

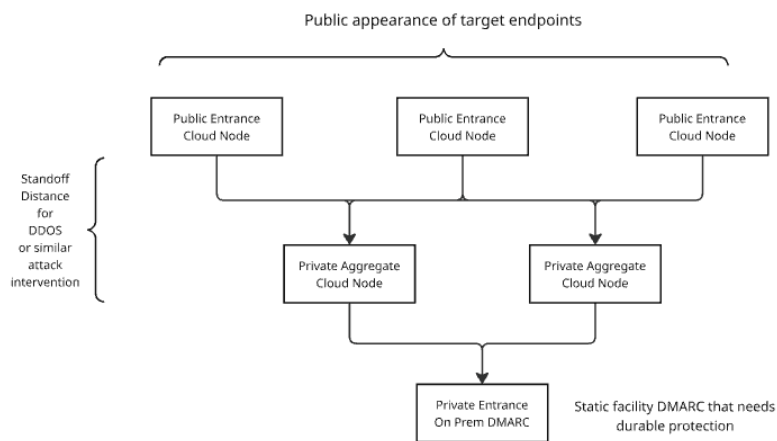


Fig. 2.1: Your caption

### 2.2.3 Gaps in the Literature

## 2.3 Methodology

### 2.3.1 Technical Research Questions

### 2.3.2 System Architecture and Design

### 2.3.3 Implementation Plan

We're starting with a Dell T610 Server. Made in 2011, this machine is about ten years old. We purchased this at a government e-waste auction. She needed some cleanup. We started on the front lawn, blowing out the dust with a leafblower.

Most of the time servers from these auctions will have minimal RAM. Usually the drives and any good accessories will be stripped out. Anytime we get a hard drive, from any source, one of the first actions we take is to erase the drives to US National Institute of Standards and Technology (NIST) 800-88 Purge[75] with dedicated devices. That kind of erasing can be done with `dd` in Kali or FreeBSD; but, the drive erasers usually have a verification feature that's helpful.

#### Hardware Configuration

This is my favorite computer. I've named her "Blondie" after Debbie Harry's New Wave band.[76]

Now, she's configured with:

- A DVD RW drive
- A LTO III[77] tape backup unit
- In her 8 drive bay, we have:
  - Two VelociRaptor[78] 80GB 10000 rpm drives.
    - ★ We'll use these in a Mirror configuration to hold the main OS.
  - Two refurb 120GB Solid State Drive (SSD)s
    - ★ We'll use this in a Mirror as ZFS log drives.
    - ★ They'll help to make writes faster.
  - Four 500GB hard drives
    - ★ We'll use this as a RAIDZ-2 array to hold the jails.
    - ★ We have some smaller Serial Attached SCSI (SAS) drives.
    - ★ We have some Serial AT Attachment (SATA) drives.
- We replaced the one CPU with two Xeon 5550[79] quad cores and added a cooling tower.
- We added more RAM. Currently, we're at 196GB.
- We added some more Network Interface Card (NIC)s.
  - We have four for the jails.
  - We have two for the OS.
  - We'll leave the two on the motherboard mostly unused, except for our iDRAC connection.
    - ★ iDRAC is a proprietary baseboard controller.

- ✱ We could use it to power cycle the machine.
- ✱ There's a web interface that let's us know about motherboard-related facts.
- ✱ We cannot give a GELI password through the console in our current setup.

- The server has two 500W power supplies.

Because the dual power supplies draw about 500 Watts apiece, this server gets its own circuit.

The drives we're using are older, but we got enough copies of the drives to be able to support backups. When making physical backups of the drives, it can be helpful to have other volumes of the same model. So, instead of getting the best available, we planned for the best we could afford at quantity.

The server does not have:

- Any type of graphics card or Graphics Processing Unit (GPU)
- Sound cards or microphone hardware
- Wireless networking or Bluetooth.

#### Intentions: Paving the Road to Jails

We'll use the box for development. It'll be our main development server. We'll use it with progRAMs that we compile on a nearby *poudriere* package building box. In those package builds, we have segregated our builds into a pattern of jails that mimic the types of jails that will be needed on most of our computers.

So, there is a jail for functional groups like: development tools, offensive and defensive security programs, browsers, desktop software, and so on. *Poudriere* jails can be filled with whatever sets of programs we like. For successful builds, we find its best to keep the jails small.

There will be three main groups of drives:

- an OS Mirror pair
- a ZFS log Mirror pair
- a ZFS RAIDZ-2 kit of four drives.

The OS pair will hold ideas like:

- jail configuration
- main host networking and Virtual Network (VNET)
- host-based user accounts and configuration.

The ZFS kits will hold disconnected jails. We'll set up a system of hierarchical jails that provide us with a development environment made up of modular jails that will be the children of an overarching parent. These jails will communicate with each other and the parent through some custom VNET. We'll do all of our jail scripting, controls, and configuration manually. Likewise, the base ZFS setup will be done with simple scripts.

#### Preparing the Baseboard

The T610 has a built-in RAID card that can't be totally bypassed. This means that since we'll want ZFS to control the drives as much as possible, we'll need to tell

the hardware RAID to buzz off. We won't want hardware RAID to participate in the process. We'll just use the software RAID through ZFS.

We can't just unplug the RAID card. We tried that, and it leads to malfunctions on this machine. So, as a compromise, what we'll tell the machine is that each drive is its own VDEV and it's RAID-0. We'll end up with as many VDEVs as we have physical drives.

To prepare for a later step in this process, we'll note the serial number of each drive we load into position, its model number, and its capacity. We'll need to know that information, and we won't want to be hassled by having to power cycle the machine in the middle of the setup process just to check on those details.

Later on, we'll set VDEVs with ZFS; but, from this point on, we'll basically ignore the RAID card's description of these devices as VDEVs.

With the baseboard prepared, and the BIOS set to allow us to boot from disc, we'll get started with a plain install of FreeBSD to the first two drives in a Mirror configuration.

### Incremental Advancement to Amnesiac System

With all of the manual coding and configuration needed to set up an amnesiac system like the one we desired, it was a big change from the normal BSD installs we had done for several years. Our first pass at building an amnesiac system decayed into manually scripting a regular operating system installation that had most of the features we desired in the amnesiac system. Both the on-disc OS system and the amnesiac system are detailed below.

### Identifying and Allocating ZFS Drive Qualities

We need to be able to detect the drives and partitions that we want to work with using `geom`, `gpart`, and `zfs` commands. If, for some reason, you can't find those drives, then you need to address that. While working with this particular computer, I've found that the leading cause of failing to detect a drive goes back to the VDEV assignment in the hardware RAID. If the drive were pulled, this particular RAID card will need to have those configuration changes addressed. Often, this means rebooting the machine. It could be done through the iDRAC baseboard controller in some models; but the one we have here is not configured to do those tasks.

What are we going to want the ZFS kit to do? How are we going to use the drives? How can we wrestle with this new type of configuration?

After reading over *FreeBSD Mastery: ZFS* by Alan Jude and Michael Lucas[107], we came to a few conclusions. We chose to use RAIDZ-2 with Log drives because of the number of drives we have[108]. Some of the topics covered in that book that we'll want to apply include ideas in Table 2.7. In our first iterations we decided to install the OS on the hard drives in a pattern that looks like a conventional installation. Once the drive partitioning and utilization commands were worked out, we then

Table 2.7: ZFS Concepts and Commands

| Topic              | Concept                                                          | Command phrase example                           |
|--------------------|------------------------------------------------------------------|--------------------------------------------------|
| Disk labels        | Use the label to associate the logical with the physical[80]     | [Host]-[PositionNumber 0-7]-[Serial_Number] [81] |
| ZFS Compression    | Faster performance: less space[82]                               | zfs set compress=gzip-6 [DIRECTORY] [83]         |
| ZFS zpool comment  | Make a note on the pool[84]                                      | zfs set comment="-" [85]                         |
| quota[86]          | Limit growth[87]                                                 | -                                                |
| reservation[88]    | Set aside space for directory and its children[89]               | -                                                |
| refreservation[90] | Set aside space but don't count snapshots and child datasets[91] | -                                                |
| canmount[92]       | Control mounting[93]                                             | zfs set canmount=off [DIRECTORY] [94]            |
| mountpoint[95]     | Specify where to mount[96]                                       | zfs set mountpoint=/SOME [DIRECTORY] [97]        |
| readonly[98]       | Control readonly[99]                                             | zfs set readonly=off [DIRECTORY] [100]           |
| exec[101]          | Control if it can execute[102]                                   | zfs set exec=off [DIRECTORY] [103]               |
| copies[104]        | Make more than one copy[105]                                     | zfs set copies=2 [DIRECTORY] [106]               |



Table 2.8: ZFS Main OS Use Hard Disk: Initial Estimates

|                                    | Estimate for 80 GB HDD |
|------------------------------------|------------------------|
| Blank: 15% "Free Space"            | 12 GB                  |
| Swap: Under-allocate, deliberately | 10 GB                  |
| ZFS Main Partition: Remainder      | 58 GB                  |

Table 2.9: ZFS Jail Use Hard Disk: Initial Estimates

|                                     | Estimate for 500 GB HDD |
|-------------------------------------|-------------------------|
| Blank: 15% "Free Space"             | 75 GB                   |
| Swap: 25% Physical RAM (196 GB max) | 50 GB                   |
| ZFS Main Partition: Remainder       | 375 GB                  |

started over and undertook the amnesiac version with the OS on a live disc that runs with attached drives.

Our goal for the system will be to split up the drives into three main groups, as described in Tables 2.8, 2.9, and 2.10. On each drive, we'll also allow:

- 15% of the disk's own capacity as "Free Space" by allocating a blank partition
- Swap by 100 | 50 | 33 | 25 percent of RAM (with a 25% minimum for all disks) or similar
- Remainder as one large ZFS main partition.

We decided we would want those qualities in an amnesiac system, with the exception of running the OS from a live disc.

For the OS-On-Disk system, this should bring us:

- Enough room for the main OS
- A 4 disk RAIDZ-2 array of almost 1.49 TB of actual storage
- Fast swap and copy-on-write
- With each disk capable of expansion or change.

The unused space on the drives (a blank partition of 15%) may look like we're grossly under-utilizing the disks. We're planning for future emergencies. Do we need more swap? Maybe we can add it in. Do we need to expand or resize the disk? Maybe we can delete that blank "Free Space" partition and use that. By not trying to use everything, we're giving ourselves some room to solve future problems.

What We Can Learn From Installing an OS

Later on, we'll install the other drives with `gpart` and `zfs` commands. This first pair, through, we could do with `bsdinstall`[109]. If we choose that, though, then we must commit the whole disk to the process. That would be how we would normally stand

Table 2.10: ZFS SLOG Log Use SSD: Initial Estimates

|                                     | Estimate for 120 GB SSD |
|-------------------------------------|-------------------------|
| Blank: 15% "Free Space"             | 18 GB                   |
| Swap: 25% Physical RAM (196 GB max) | 50 GB                   |
| ZFS Log Partition: Remainder        | 52 GB                   |

up a disk drive. In this case, we'll conduct a manual process that mimics the main steps of `bsdinstall`<sup>[110]</sup> to show that the installation can be done manually. Part of these same kinds of steps are critical tasks when administrating and installing jails.

Basic Operating system Installation Goals

We're going to do an early phase, basic installation of the OS. At this point, we'll really only set up a pair of users and some plain vanilla settings for the OS. We want to be able to:

- to establish our ZFS kit
- to configure our main jail relationships
- to be able to check our networking with other hosts
- to set the basic operating system to call packages from our package server.

We can come back later and make a more sophisticated environment. At this stage, we will just lay the foundation for that work. If we don't get the ZFS kit stood up, then our more complex work with jails won't have an environment in which to thrive.

Main Steps for FreeBSD Installation from `bsdinstall`

<sup>[111]</sup>

We can see that `bsdinstall` <sup>[112]</sup> is broken up into about a dozen main segments. They are:

- Keyboard mapping
- Hostname
- Networking configuration
- Wireless networking
- Partitioning
- Mounting temporary root
- Fetch the OS
- Extract the OS
- Set a root password
- Set the time
- Add users
- Enable some services
- Copy in the config <sup>[2]</sup>.

We're not going to do all of these. We'll start with partitioning. We won't do

any wireless networking or set a root password. We're going to skip the wireless networking for now because the hardware doesn't have that capability. In the newer versions of `bsdinstall`, there are some protective measures not listed above. We won't tackle those just yet.

### No Root Password

We won't set a root password because we will expect to always Secure Shell (SSH) into the root account with a key. By not setting a root password, we will create a situation where a key is always required. This can be dangerous in that it can create a single point of failure for the system.

Another risk is that we will require the ability to SSH in as root. Later, as we establish software firewalls on the host, we will want to limit what addresses can be used to access this root with SSH. It's a little dangerous, but it's also a little different.

We have some risk controls that will be in place. First, we often store our keys in a special collection of USB drives. Next, we will have 2FA on the SSH that will require a TOTP (Time-based One Time Password). Finally, we will limit what's done in this drive. Since most of our programs and work will be done in the jails, there is a limited amount of influence that this part of the system will have. It's there to start the box and spin up the jails. If we ever have to totally destroy this installation and hook up our jails to another system, that is possible.

What do we gain from no root password? An extra measure of security. If there is no password, then password enumeration attempts should all fail. This not only goes for root, but it goes for other users. Many of the automated login attacks we've seen have been simple dictionary attacks; these brute force exercises rarely attempt to account for the types of login procedures that we expect to put in place.

Not having a root password may also truncate or stop our ability to use commands like `su` or `sudo`. As we're trying to stop privilege escalation, we'll also be stopping legitimate means of `priv-esc`. It could be inconvenient, but we've used this technique before.

Also, there is another risk: leaving the needed key lying around. For example, if we give the key to a user account operator, like ourselves, and then we import copies of the key to the box for easy use: well, then those keys could be used and re-used locally. This would be the equivalent for storing a plain-text password on the box. They keys are ultimately simple text files with decorated, long, strings of hashes. It's the software that goes with the keys, combined with the configuration of the programs for logging in to a service, that makes them special. Switching from passwords to keys is not a perfect answer; but, it's the answer we'll use this time.

If someone wishes to put a root password in place, then that's a procedure that will be easy to implement. For now, though, there will be no root password. And that means we have to build the keys we will use to log in.

### Be Able to Build the Keys

We can build our keys using the live disc's shell. and a thumbdrive. The main command to build the keys is:

```
ssh-keygen -b 1024 -t rsa[113]
```

Copy the files output to a thumbdrive and keep them handy for mounting.

#### Partition, Mount, and Use a Thumbdrive from Live Disc's Shell

Let's take some time to go over some detailed instructions on how to build those keys and get them onto a thumbdrive. The live disc's shell mostly uses a read-only file system. So, some people may not be familiar with how we are going to get the keys made and saved anywhere that will persist.

#### Start the Box and the Live Disc's Shell

With the hardware booted to the LiveCD (installation disc), we select "Shell" from the three main choices in the first menu. This drops us into the shell. If we try to save a file of any kind, we'll probably get an error message back that says we're in a read-only file system. Our main set of directions for this is in the FreeBSD Handbook. We look carefully at the sections for USB storage and "Adding Disks"

#### Be Able to Plug in the USB Thumbdrive and Find Its Logical Address

```
camcontrol devlist[114]
```

This should show the USB drive. If the list is long, we can pipe it to more and slowly advance through the list. In our case, the USB was at da0.

The listing should also show any other storage device that's plugged in to the machine. So, all of those hard drives loaded into the bays should appear. If they don't, then take it as a warning sign that we're having a problem with the RAID card's physical drive VDEVs. On the T610, we were seeing drives in positions 0 to 7 in addition to the USB thumbdrive we're working on now.

#### Be Able to Partition the USB

Since our USB has not been prepared for use with anything, we have to partition it. We can pretty much use the default commands from 17.2 "Adding Disks" to set up the drive. Here, we create a simple UFS file system and mount it to a directory that we'll put in /tmp. If our USB is already partitioned, we would want to skip this part.

```
gpart create -s GPT da0[115]  
gpart add -t freebsd-ufs -a 1M da0[116]  
newfs -U /dev/da0p1[117]
```

#### Mount the USB to the /tmp Directory:

```
mkdir /tmp/someDir[118]
```

```
mount /dev/da0p1 /tmp/someDir[119]
```

Be Able to Write to the USB

We could now write a simple text file onto /tmp/someDir, unmount the drive, and then re-mount it to check our work. That would go like:

```
cd /tmp[120]
umount /tmp/someDir[121]
rm -r /tmp/someDir[122]
```

Unplug the drive. Plug it in another slot. Use `camcontrol devlist` to find it again. It'll probably get the same address, like "da0".

```
mkdir /tmp/otherDir[123]
mount /dev/da0p1 /tmp/otherDir[124]
ls /tmp/otherDir[125]
```

Be Able to Put Some Key Pairs on the USB

Using these same principles, we can mount our USB thumbdrive, generate SSH keys, and save them there for later.

```
mkdir /tmp/otherDir/keys[126]
ssh-keygen -b 1024 -t rsa[127]
```

During the `ssh-keygen` dialog, simply direct the program to save the keys to /tmp/otherDir/keys. Be sure to record your passphrase somewhere.

Be Able to Use Commands to Build Out the Drives

We'd like to show a quick overview of some of the drive-related commands. These will be applied soon.

To list what devices are detected by the OS, we can use:

```
gpart show[128]
```

We will want to note the serial number of each drive, by position. We then note which drives we want to use for the main body of the RAIDZ-2 kit. All of this will become one pool of ZFS VDEVs. Then we'll add in the log mirrors. So, in the sequence of 0-7, numbering the drives, we should expect to skip those two for now.

Make the notes if you haven't already because we're going to list all of the drives for the main pool in one pop. This task may have been done before we set up the physical VDEVs for the hardware RAID.

We're going to choose to put a swap partition on the SSDs. We can add one onto the other disks, but a maximum of 4 swap partitions is recommended in the FreeBSD Handbook. Later on, we will be able to select swap partitions by mounting them and using commands like `swapon` `swapoff`.

To put the ZFS partition on the remaining disks:

```
gpart create -s SCHEME DISK_DEVICE[129]
```

```
gpart create -s gpt da4[130]
```

To add the main ZFS partition:

```
gpart add -a 1m -l ZFS_NUMBER \  
-t freebsd-zfs DISK_DEVICE[131]
```

```
gpart add -a 1m -l ZFS-BLONDIE-4-WD123465789 \  
-t freebsd-zfs da4[132]
```

To add the partition's swap:

```
gpart add -a 1m -slg -l SWAP_NUMBER \  
-t freebsd-swap DISK_DEVICE[133]
```

```
gpart add -a 1m -slg -l SWAP-BLONDIE-4-WD123465789 \  
-t freebsd-swap da4[134]
```

To add a logging partition on the SLOG SSDs:

```
gpart add -a 1m -l ZLOG_NUMBER \  
-t freebsd-zfs DISK_DEVICE[135]
```

```
gpart add -a 1m -l ZLOG-BLONDIE-2-WD123465789 \  
-t freebsd-zfs da2[136]
```

With each disk partitioned, we can add them to the pool for our RAIDZ-2 kit.

```
zpool create RAID POOL_NAME DISK_LABEL/ZFS_PARTITION_LABEL  
...next disk/partition...[137]
```

```
zpool create BLONDIE-MAIN raidz2 \[138]  
BLONDIE-4-WD12345/ZFS-BLONDIE-4-WD12345 \  
BLONDIE-5-WD12345/ZFS-BLONDIE-5-WD12345 \  

```

```
BLONDIE-6-WD12345/ZFS-BLONDIE-6-WD12345 \
BLONDIE-7-WD12345/ZFS-BLONDIE-7-WD12345
```

We can add on the mirrors for logging:

```
zpool add BLONDIE-MAIN log mirror \[139]
BLONDIE-2-WD12345/ZLOG-BLONDIE-2-WD12345 \
BLONDIE-3-WD12345/ZLOG-BLONDIE-3-WD12345
```

Once we get to the point that we have an operating system on disks, and disks ready for use with logs and jails, then we're at the end of this phase of system administration.

### Goals for Amnesiac Systems

We chose at this point to do some experimental installs. For clarity, notes on those are not presented here. We did change our goals for the system. We recognized that we wanted:

- Use a LiveCD for an immutable OS
- Encrypt disc partitions instead of the whole disc
- Ensure we save keys and metadata to promote disc recovery
- Prepare commands in advance to reduce complexity.

From our testing, we recognize that a Live CD offered the best answer for an immutable system. It does come at the price of some instability. The "El Torrito" discs have a limited set of services. Adjusting the configuration of the operating system means drafting, burning, and testing a new disc. The "shell" instance on a FreeBSD installation disc is reportedly less stable; but, we did not know why or how it was less stable than a normal operating system. We worked through our disc commands and kept careful notes. Adding the physical serial number of the disc alone increased the complexity of manually typing in each command. We discovered the need to encrypt the discs by partitions instead of the whole disc because there was a visibility problem during the execution of discovering and attaching the discs. There was a poorly documented detail about protecting the disc metadata and keys. The keys are provided during the setup dialog; but, these highly important items will disappear after the setup process closes. This means it is important that an operator recognize the need to intercept and to record these keys. Without them, a later recovery of a disc is impossible. Since one typo would derail our experimental setups, we had to keep a second support computer on hand to record notes and consult references.

As our experimentation concluded, we had learned enough to stand a reasonable chance to setup an amnesiac machine. Its features include:

- An immutable OS
- Encrypted disc partitions
- Swap and logging for ZFS disc features
- Metadata and keys saved to a separate USB drive

Table 2.11: FreeBSD Swap Disk – Estimate of 500 GB HDD

|                          |        |
|--------------------------|--------|
| GEOM Disk List Mediasize | 465 GB |
| Blank: 15% "Free Space"  | 70 GB  |
| Swap                     | 395 GB |

Table 2.12: ZFS Cache (Read Cache) Disk – Estimate of 80 GB HDD

|                          |       |
|--------------------------|-------|
| GEOM Disk List Mediasize | 74 GB |
| Blank: 15% "Free Space"  | 11 GB |
| Swap                     | 63 GB |

- A disc for holding FreeBSD jails (containerization)
- Enough room to expand, to update, and to recover.

Configure for SWAP Disk, ZFS Cache, ZFS SLOG Mirror, and ZFS RAIDz2

Changed config to keep the RAIDZ2 for main work. I was having trouble with the SSDs; so, I removed them and replaced them with two 80GB Western Digital Velociraptors. I added another WD Velociraptor for cache. Then we added a Western Digital Blue 500GB for a swap disk. That sounds like a lot of swap, but we hope to upgrade to near 196GB of RAM this year. So, a little more than twice that would be 400GB of swap. The next disk size on hand was 500GB.

We had some mis-aligned partitions with RAIDZ2 work for Blondie. We were using the full 500GB on the label as our actual amount of media to work with. Since less is available, we will have to set those up again.

We used gpart delete to delete each partition. Then we used gpart destroy to delete the partition table for the whole disk. In the end, gpart showed us only the CD.

Table 2.13: ZFS SLOG Log Mirror (Write Cache) Disk – Initial Estimates

|                                     |                         |
|-------------------------------------|-------------------------|
|                                     | Estimate for 120 GB SSD |
| GEOM Disk List Mediasize            | 465 GB                  |
| Blank: 15% "Free Space"             | 18 GB                   |
| Swap: 25% Physical RAM (196 GB max) | 50 GB                   |
| ZFS Log Partition: Remainder        | 52 GB                   |



Table 2.14: ZFS Jail Use Hard Disk – Estimate of 500 GB HDD

|                                     | Estimate for 500 GB HDD |
|-------------------------------------|-------------------------|
| GEOM Disk List Mediasize            | 465 GB                  |
| Blank: 15% "Free Space"             | 75 GB                   |
| Swap: 25% Physical RAM (196 GB max) | 50 GB                   |
| ZFS Main Partition: Remainder       | 375 GB                  |

Startup with **bootonly.iso**

For this run, we'll begin by using the boot-only disk. We won't plan to ever boot from these hard drives. We'll just use the shell on the CD. As with the others, we'll mount a USB to store metadata information. We'll partition and encrypt the disks. In the following examples, we provide the actual serial numbers for the hard drives that were used. This example was derived from notes we maintained during experimentation; serial numbers in other instances will vary, of course.

Mount the USB to Save the Metadata Backup File

```
mkdir /tmp/usb3[140]
mount /dev/da0p1 /tmp/usb3[141]
```

Create Swap Disk in Bay 0

```
gpart create -s GPT mfid0[142]
gpart add -a 5K -s 395G -t freebsd-swap \
-l ZFS-BLONDIE-0-SWAP-WCC2ERZ56512 mfid0[143]
```

Create ZFS Cache in Bay 1

```
gpart create -s GPT mfid1[144]

gpart add -a 5K -s 63G -t freebsd-zfs \
-l ZFS-BLONDIE-1-CACHE-WXL1E40C1352 mfid1[145]
```

Create ZFS SLOG in Bays 2 and 3

```
gpart create -s GPT mfid2[146]

gpart add -a 5K -s 63G -t freebsd-zfs \
-l ZFS-BLONDIE-2-SLOG-WXL1E4067503 mfid2[147]

gpart create -s GPT mfid3[148]
```

```
gpart add -a 5K -s 63G -t freebsd-zfs \  
-l ZFS-BLONDIE-3-SLOG-WXC0CA9V8431 mfid3[149]
```

### Create ZFS Work Disks for Jails

#### Bay 4

```
gpart create -s GPT mfid4[150]
```

```
gpart add -a 5K -s 50G -t freebsd-swap \  
-l ZFS-BLONDIE-4-SWAP-WCC2EE134983 mfid4[151]
```

```
gpart add -s 345G -t freebsd-zfs \  
-l ZFS-BLONDIE-4-FILES-WCC2EE134983 mfid4[152]
```

#### Bay 5

```
gpart create -s GPT mfid5[153]
```

```
gpart add -a 5K -s 50G -t freebsd-swap \  
-l ZFS-BLONDIE-5-SWAP-WMC2E3914420 mfid5[154]
```

```
gpart add -s 345G -t freebsd-zfs \  
-l ZFS-BLONDIE-5-FILES-WMC2E3914420 mfid5[155]
```

#### Bay 6

```
gpart create -s GPT mfid6[156]
```

```
gpart add -a 5K -s 50G -t freebsd-swap \  
-l ZFS-BLONDIE-6-SWAP-WCC2ETT39577 mfid6[157]
```

```
gpart add -s 345G -t freebsd-zfs \  
-l ZFS-BLONDIE-6-FILES-WCC2ETT39577 mfid6[158]
```

#### Bay 7

```
gpart create -s GPT mfid7[159]
```

```
gpart add -a 5K -s 50G -t freebsd-swap \  
-l ZFS-BLONDIE-7-SWAP-WCAYU2878968 mfid7[160]
```

```
gpart add -s 345G -t freebsd-zfs \  
-l ZFS-BLONDIE-7-FILES-WCAYU2878968 mfid7[161]
```

We should note that “Bay 7” is an observation of the physical hardware. The device (“mfid7”) could have whatever number assigned at the end that the machine gave it. If the hardware RAID had a problem with recognizing a device, the number might not correlate to what bay it is in. When powering up, check your geometry.

Encrypt those partitions

During this phase, we’ll encrypt the partitions. That means that we should have some recordkeeping materials on hand. Use a password manager or a notebook to carefully record the password that is about to be typed in. With each one completed, the machine will tell us that it has saved a file of backup metadata. We will want to copy that metadata file to the thumbdrive. If the encryption gets jammed up, or if there is some type of problem, then the metadata file will be used as part of the recovery process. Without a metadata file, recovering an encrypted partition can be difficult or impossible.

If you lose the metadata file, but still have routine access to the geli encryption because you know the password and the machine is functioning alright, then you can export another copy of the metadata. This means that anyone with root access to the machine, who can use geli commands, would also be able to export such a file. Keep this in mind if you need to secure your machine.

When we give the subcommand to “init,” the machine will ask us to give the new password twice. This is part of the password setting process. When we give the subcommand to “attach,” the machine will ask us to give the password as part of the normal attachment process. Type carefully, and look at the records carefully, because this is the main chance to make sure the password notes are right.

Commands to Encrypt the Partitions

```
geli init -l 256 mfid0p1[162]
geli attach mfid0p1[163]
```

```
geli init -l 256 mfid1p1[164]
geli attach mfid1p1[165]
```

```
geli init -l 256 mfid2p1[166]
geli attach mfid2p1[167]
```

```
geli init -l 256 mfid3p1[168]
geli attach mfid3p1[169]
```

```
geli init -l 256 mfid4p1[170]
geli attach mfid4p1[171]
```

```
geli init -l 256 mfid4p2[172]
geli attach mfid4p2[173]
```

```
geli init -l 256 mfid5p1[174]
geli attach mfid5p1[175]

geli init -l 256 mfid5p2[176]
geli attach mfid5p2[177]

geli init -l 256 mfid6p1[178]
geli attach mfid6p1[179]

geli init -l 256 mfid6p2[180]
geli attach mfid6p2[181]

geli init -l 256 mfid7p1[182]
geli attach mfid7p1[183]

geli init -l 256 mfid7p2[184]
geli attach mfid7p2[185]
```

### To Preserve the Metadata Backup Files

Let us reiterate: preserving these backups at the time of disk creation is essential for later recoveries. If these files are not saved, then a powerdown of the system will cause them to be lost. They're automatically stored in an ephemeral directory. Conserve the ability to recover a disk later by recording these metadata backup files to a safe location.

Each metadata backup file was probably created with a name that does not look like the drive's label. Instead, the machine assigns a backup file name based on geometry. Since we can have a hardware configuration change that could mess up that pattern, we will want to save our backup files with names that look more like the disk labels. That way, if the drives move, then we can better discover which metadata file goes with it.

```
cp /var/backups/mfid0p1.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-0-SWAP-WCC2ERZ56512.eli[186]

cp /var/backups/mfid1p1.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-1-CACHE-WXL1E40C1352.eli[187]

cp /var/backups/mfid2p1.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-2-SLOG-WXL1E4067503.eli[188]

cp /var/backups/mfid3p1.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-3-SLOG-WXC0CA9V8431.eli[189]
```

```

cp /var/backups/mfid4p1.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-4-SWAP-WCC2EE134983.eli[190]

cp /var/backups/mfid4p2.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-4-FILES-WCC2EE134983.eli[191]

cp /var/backups/mfid5p1.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-5-SWAP-WMC2E3914420.eli[192]

cp /var/backups/mfid5p2.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-5-FILES-WMC2E3914420.eli[193]

cp /var/backups/mfid6p1.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-6-SWAP-WCC2ETT39577.eli[194]

cp /var/backups/mfid6p2.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-6-FILES-WCC2ETT39577.eli[195]

cp /var/backups/mfid7p1.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-7-SWAP-WCAYU2878968.eli[196]

cp /var/backups/mfid7p2.eli /tmp/usb3/backups
_27SEP2020/ZFS-BLONDIE-7-FILES-WCAYU2878968.eli[197]

```

### File Name Check: Editing Labels

We got through all that, and found that we had named a file incorrectly. During experimentation, we had a typo; BAY 4 should be WCC2EE134983 but that's not what we saw when we checked our work. Not the end of the world. We can move the .eli metadata file, and we can use `gpart`[198] modify to change the label.

```

gpart modify -i 1 \
-l ZFS-BLONDIE-4-SWAP-WCC2EE134983 mfid4[199]

gpart modify -i 2 \
-l ZFS-BLONDIE-4-FILES-WCC2EE134983 mfid4[200]

```

`gpart show`[201] and `geli status`[202] should show our drives as we expect them to be.

At this point, all of the drives should be partitioned and decrypted and attached. We can build a zpool for our ZFS file system. We can attach that swap drive for the benefit of our operating system right now.

To Use the Swap Disk

```
swapon mfid0p1.eli[203] swapinfo -k[204]
```

64G turned out to be the maximum capacity. We will need to fix up the swap better. We can fit 6 x 64G (384G) of swap partitions in our planned space. So, let's modify and add.

```
gpart resize -i 1 -s 64G mfid0[205]
```

We would need to add, to init and to attach five more swap partitions. We also have to restore mfid0 because there's a problem with reading the metadata.

```
gpart recover mfid0[206]  
gpart status mfid0[207]  
geli attach mfid0p1[208]
```

```
gpart add -s 64G -t freebsd-swap \  
-l ZFS-BLONDIE-0-SWAP-B-WCC2ERZ56512 mfid0[209]
```

```
gpart add -s 64G -t freebsd-swap \  
-l ZFS-BLONDIE-0-SWAP-C-WCC2ERZ56512 mfid0[210]
```

```
gpart add -s 64G -t freebsd-swap \  
-l ZFS-BLONDIE-0-SWAP-D-WCC2ERZ56512 mfid0[211]
```

```
gpart add -s 64G -t freebsd-swap \  
-l ZFS-BLONDIE-0-SWAP-E-WCC2ERZ56512 mfid0[212]
```

```
gpart add -s 64G -t freebsd-swap \  
-l ZFS-BLONDIE-0-SWAP-F-WCC2ERZ56512 mfid0[213]
```

Then we need to establish encryption for all of those.

```
geli init -l 256 mfid0p2[214]  
geli attach mfid0p2[215]  
geli init -l 256 mfid0p3[216]  
geli attach mfid0p3[217]  
geli init -l 256 mfid0p4[218]  
geli attach mfid0p4[219]  
geli init -l 256 mfid0p5[220]  
geli attach mfid0p5[221]  
geli init -l 256 mfid0p6[222]  
geli attach mfid0p6[223]
```

Using `swapinfo`[224], `swapon`[225] and `swapoff`[226], we were able to remove some of the other swap partitions and add on these. Did see a kernel limit on swap. Total configured swap exceeded `kern.maxswzone` (98087976 pages). We could tune the kernel, or we could reduce swap. For now, we reduce swap from 384GB to 324GB.

Simiarily, we could add on the other swap partitions, although in our current use, we probably won't need them. They would serve us better in cases where we were doing more intensive work, or working on another machine. We can mount them now, anyway, to see that it can be done.

To add on all the other swap partitions:

```
swapon /dev/mfid4p1.eli[227]
swapon /dev/mfid5p1.eli[228]
swapon /dev/mfid6p1.eli[229]
swapon /dev/mfid7p1.eli[230]
swapinfo -h[231]
```

Create the zpool for BLONDIE

We'll now create a mountpoint on the tmpfs, create the zpool, and then add in the cache and SLOG Mirrored disks.

Create a mountpoint

Create a mountpoint so that we will be able to mount the zpool without errors as we create it.

```
mkdir /tmp/BLONDIE[232]
```

## 2.4 Analysis

### 2.4.1 Quality

### 2.4.2 Tests

## 2.5 Discussion

## 2.6 Chapter References

### 2.6.1 Books

For our ZFS planning, we read: Lucas, Michael W, and Allan Jude. *FreeBSD Mastery: ZFS*. Tilted Windmill Press, 21 May 2015.

## 2.6.2 MAN Pages

Online MAN pages were available at: [man.freebsd.org](http://man.freebsd.org) with an interface that allows for keyword based searches.

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